

Discrete Structures and Theory of Logic

Unit 1: Set Theory, Relations, Posets and Lattices

Complete Unit Notes

Comprehensive classroom and self-study material for second-year B.Tech CSE students, aligned with the autonomous syllabus.

Prepared By

Aditya Tandon

Department of Computer Science and Engineering

Academic Session 2026–27

Contents

Syllabus Coverage	v
1 Introduction to Set Theory	1
1.1 Why Set Theory Is Important	1
1.2 Well-Defined Collections	2
1.3 Methods of Representing Sets	2
1.3.1 Roster or Tabular Form	2
1.3.2 Set-Builder Form	2
1.3.3 Descriptive Form	2
1.4 Important Number Sets	2
1.5 Types of Sets	3
1.5.1 Empty or Null Set	3
1.5.2 Singleton Set	3
1.5.3 Finite and Infinite Sets	3
1.5.4 Equal Sets	3
1.5.5 Equivalent Sets	3
1.5.6 Universal Set	3
1.6 Subsets and Proper Subsets	4
1.7 Proper Subset	5
1.8 Difference between Subset and Proper Subset	5
1.9 Important Facts	5
1.10 Cardinality	7
1.11 Power Set	7
2 Set Operations and Algebra of Sets	8
2.1 Union	8
2.2 Intersection	8
2.3 Difference	8
2.4 Complement	8
2.5 Symmetric Difference	9
2.6 Venn Diagram Interpretation	9
2.7 Fundamental Laws of Set Algebra	9
2.8 Proof of a Set Identity by Element Method	9
2.9 Duality Principle	10
2.10 Computer Science Interpretation	10
2.11 Solved Problems	10
3 Cartesian Product	12
3.1 Ordered Pairs	12
3.2 Cardinality of a Cartesian Product	12
3.3 Cartesian Powers	12
3.4 Properties of Cartesian Products	13
3.5 Matrix View	13
3.6 Applications in Computer Science	13

4	Inclusion–Exclusion Principle	14
4.1	Why Direct Addition Fails	14
4.2	Exactly One and Neither	14
4.3	Three-Set Formula	14
4.4	Finding Region-Wise Counts	15
4.5	Divisibility Problem	15
5	Fuzzy Sets	17
5.1	Introduction	17
5.2	Need for Fuzzy Sets	17
5.3	What is a Fuzzy Set?	18
5.4	Example of a Fuzzy Set	18
5.5	Crisp Set vs Fuzzy Set	19
5.6	Basic Operations on Fuzzy Sets	19
5.6.1	Union	19
5.6.2	Intersection	19
5.6.3	Complement	20
5.7	Applications of Fuzzy Sets	20
5.8	Solved Example	20
5.9	Key Points to Remember	20
5.10	Practice Questions	21
6	Multisets	22
6.1	Introduction	22
6.2	Why Do We Need Multisets?	22
6.3	Definition	23
6.4	Representation of a Multiset	23
6.5	Another Example	23
6.6	Difference between Set and Multiset	23
6.7	Basic Operations on Multisets	23
6.8	Applications of Multisets	24
6.9	Solved Example	25
6.10	Multiset vs Fuzzy Set	25
6.11	Practice Questions	25
7	Binary Relations and Their Representation	27
7.1	Motivation	27
7.2	Domain, Codomain and Range	27
7.3	Number of Relations	28
7.4	Commonly Defined Relations	28
7.4.1	Relation Defined by a Formula	28
7.4.2	Divisibility Relation	28
7.5	Representation by Ordered Pairs	28
7.6	Arrow Diagram	28
7.7	Zero–One Matrix Representation	29
7.8	Directed Graph Representation	29
7.9	Converting Between Representations	30
7.10	Special Relations Introduced at the Representation Stage	30
7.11	Relation versus Function	30
7.12	Computer Science Applications	30
7.13	Solved Problems	31

8	Revision Exercises for the Completed Topics	32
8.1	Objective Questions	32
8.2	Short-Answer Questions	32
8.3	Long-Answer and Numerical Questions	33
9	Properties and Types of Relations	34
9.1	Reflexive and Irreflexive Relations	34
9.2	Symmetric, Asymmetric and Antisymmetric Relations	34
9.3	Transitive Relations	34
9.4	Testing from a Matrix	34
9.5	Testing from a Digraph	35
9.6	Common Types of Relations	35
9.7	Property Comparison Table	35
9.8	Solved Classification Problems	35
10	Equality, Inverse and Composite Relations	37
10.1	Equality of Relations	37
10.2	Inverse Relation	37
10.3	Composite Relations	37
10.4	Matrix Method for Composition	37
10.5	Powers of a Relation	37
11	Recursive Definition of Relations	38
11.1	General Form	38
11.2	Recursive Relations in Computer Science	38
12	Equivalence Relations and Partitions	39
12.1	Equivalence Classes	39
12.2	Congruence Modulo n	39
12.3	Partitions	39
12.4	Quotient Set	40
13	Closure of Relations	41
13.1	Reflexive Closure	41
13.2	Symmetric Closure	41
13.3	Transitive Closure	41
13.4	Applications	41
14	Warshall's Algorithm	42
14.1	Purpose	42
14.2	Algorithm	42
14.3	Detailed Example	42
14.4	Complexity	43
14.5	C++ Implementation	43
15	Partially Ordered Sets	44
15.1	Standard Examples	44
15.2	Comparable and Incomparable Elements	44
15.3	Minimal, Maximal, Least and Greatest Elements	44
15.4	Upper and Lower Bounds	45

16 Hasse Diagrams	46
16.1 Purpose	46
16.2 Construction Procedure	46
16.3 Example: Divisors of 12	46
16.4 Example: Power Set	46
16.5 Reading Information from a Hasse Diagram	47
17 Lattices	48
17.1 Algebraic Laws of Lattices	48
17.2 Examples of Lattices	48
17.3 Bounded Lattice	48
17.4 Complemented Lattice	48
17.5 Distributive Lattice	48
17.6 Modular Lattice	49
17.7 Complete Lattice	49
17.8 Hierarchy of Important Classes	49
17.9 Testing Whether a Poset Is a Lattice	49
17.10 Solved Example: Divisors of 30	49
17.11 Counterexample	49
17.12 Applications	49
18 Lexicographic Ordering	50
18.1 Strings	50
18.2 Applications	50
19 Integrated Solved Problems	51
20 Common Errors and Correction Strategy	52
21 Practice Questions	53
21.1 Multiple-Choice Questions	53
21.2 Short-Answer Questions	54
21.3 Long-Answer and Numerical Questions	55
22 Rapid Revision Sheet	56

Syllabus Coverage

This document comprehensively covers all topics of **Unit 1: Set Theory, Relations, POSETS and Lattices** as prescribed in the autonomous syllabus. The contents are organized from fundamental concepts to advanced topics, including definitions, theorems, proofs, solved examples, applications, and practice problems.

- | | |
|---|---------------------------------------|
| 1. Introduction to Discrete Structures | 17. Equality of Relations |
| 2. Sets and Set Operations | 18. Recursive Definition of Relations |
| 3. Types of Sets | 19. Equivalence Relations |
| 4. Cardinality of Sets | 20. Partitions |
| 5. Power Sets | 21. Closure of Relations |
| 6. Algebra of Sets | 22. Warshall's Algorithm |
| 7. De Morgan's Laws | 23. Partially Ordered Sets (Posets) |
| 8. Venn Diagrams | 24. Totally Ordered Sets |
| 9. Cartesian Product | 25. Well Ordered Sets |
| 10. Inclusion–Exclusion Principle | 26. Hasse Diagrams |
| 11. Binary Relations and their Representation | 27. Bounded Lattices |
| 12. Fuzzy Sets | 28. Complemented Lattices |
| 13. Multisets | 29. Modular Lattices |
| 14. Properties of Relations | 30. Distributive Lattices |
| 15. Types of Relations | 31. Complete Lattices |
| 16. Composite Relations | 32. Lexicographic Ordering |

1 Introduction to Set Theory

Learning Outcomes

After completing this section, students should be able to explain the role of set theory in computer science, use standard set notation, represent sets in different forms, identify common types of sets, and determine cardinality and power sets.

1.1 Why Set Theory Is Important

Set theory provides a precise language for describing collections of objects. Nearly every area of computer science uses sets directly or indirectly. Examples include:

- a database table as a set of records;
- the alphabet of a programming language as a finite set of symbols;
- the states of a finite automaton as a set;
- the vertices and edges of a graph as sets;
- the solution space of a computational problem as a set;
- permissions assigned to users as sets of access rights.

Basic Idea

A **set** is a well-defined collection of distinct objects. The objects belonging to a set are called its **elements** or **members**.

Definition 1.1 (Set). A set is a well-defined collection of distinct objects considered as a single mathematical entity.

If A is a set and x is an element of A , we write

$$x \in A.$$

If x does not belong to A , we write

$$x \notin A.$$

Example 1.1. Let

$$A = \{2, 4, 6, 8\}.$$

Then $4 \in A$, whereas $5 \notin A$.

Common Mistake

A set contains distinct elements. Therefore,

$$\{1, 1, 2, 2, 3\} = \{1, 2, 3\}.$$

Repetition does not change an ordinary set.

1.2 Well-Defined Collections

A collection is well-defined when there is an unambiguous rule for deciding whether a given object belongs to it.

Example 1.2. The collection of all even natural numbers less than 20 is well-defined:

$$E = \{2, 4, 6, 8, 10, 12, 14, 16, 18\}.$$

By contrast, “the set of intelligent students” is not well-defined unless a measurable criterion for intelligence is specified.

1.3 Methods of Representing Sets

1.3.1 Roster or Tabular Form

All elements are listed inside braces.

$$V = \{a, e, i, o, u\}.$$

This method is suitable for small finite sets.

1.3.2 Set-Builder Form

A property satisfied by the elements is stated.

$$E = \{x \in \mathbb{N} \mid x < 10 \text{ and } x \text{ is even}\}.$$

The vertical bar $\mid$ is read as “such that.”

Example 1.3. Convert

$$A = \{1, 4, 9, 16, 25\}$$

into set-builder form.

Solution. Each element is a square of a natural number from 1 to 5. Hence,

$$A = \{x^2 \mid x \in \mathbb{N}, 1 \leq x \leq 5\}.$$

1.3.3 Descriptive Form

The set is described verbally.

For example, P may denote the set of prime numbers less than 20.

1.4 Important Number Sets

The following standard notation is used throughout discrete mathematics:

$\mathbb{N} = \{1, 2, 3, \dots\}$	natural numbers,
$\mathbb{W} = \{0, 1, 2, 3, \dots\}$	whole numbers,
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	integers,
$\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$	rational numbers,
$\mathbb{R}$	the set of real numbers.

1.5 Types of Sets

1.5.1 Empty or Null Set

Definition 1.2. A set containing no element is called the empty set and is denoted by $\emptyset$ or $\{\}$.

Example 1.4.

$$A = \{x \in \mathbb{N} \mid x < 0\} = \emptyset.$$

1.5.2 Singleton Set

A set containing exactly one element is called a singleton set.

$$A = \{0\}.$$

Note that $\emptyset \neq \{\emptyset\}$. The first set has no element, whereas the second contains one element, namely the empty set.

1.5.3 Finite and Infinite Sets

A finite set contains a countable and limited number of elements. An infinite set does not.

$$A = \{1, 2, 3, 4\} \quad \text{is finite,}$$

whereas $\mathbb{N}$ is infinite.

1.5.4 Equal Sets

Definition 1.3. Two sets A and B are equal if they contain exactly the same elements:

$$A = B \iff \forall x (x \in A \leftrightarrow x \in B).$$

The order of elements is irrelevant:

$$\{1, 2, 3\} = \{3, 2, 1\}.$$

1.5.5 Equivalent Sets

Two finite sets are equivalent if they have the same number of elements, even if their elements are different.

$$A \sim B \iff |A| = |B|.$$

For example, $\{1, 2, 3\}$ and $\{a, b, c\}$ are equivalent but not equal.

1.5.6 Universal Set

The universal set U contains all objects currently under consideration. Complements are always taken relative to a specified universal set.

1.6 Subsets and Proper Subsets

Before understanding subsets, let us consider a simple real-life example.

Suppose a class has the following students

$$\{\text{Aman, Ravi, Priya, Neha, Karan}\}$$

Now consider another group containing only

$$\{\text{Aman, Ravi}\}$$

Observe that every student in the second group is already present in the first group.

Therefore, the second group is said to be a **subset** of the first group.

In simple words,

Remember

A set is called a subset of another set if **every element of the first set is also present in the second set.**

Definition 1.4 (Subset). A set A is said to be a **subset** of another set B if every element of A belongs to B .

It is written as

$$A \subseteq B$$

and is read as “**A is a subset of B.**”

Mathematically,

$$A \subseteq B \iff \forall x, x \in A \Rightarrow x \in B$$

Example 1

Let

$$A = \{1, 2\}$$

and

$$B = \{1, 2, 3, 4\}$$

Since every element of A is present in B ,

$$A \subseteq B$$

is true.

Example 2

Let

$$A = \{2, 4, 6\}$$

and

$$B = \{1, 2, 3, 4, 5\}$$

Here, 6 is present in A but not in B .

Hence,

$$A \not\subseteq B$$

Therefore, A is **not** a subset of B .

1.7 Proper Subset

Now suppose

$$A = \{1, 2\}$$

and

$$B = \{1, 2, 3\}$$

Notice two things.

- Every element of A is present in B .
- B has one extra element (3).

Therefore,

A is not only a subset, but also a **proper subset** of B .

Definition 1.5 (Proper Subset). A set A is called a **proper subset** of B is

- every element of A is present in B , and
- A and B are not equal.

It is written as

$$A \subset B$$

Example

Let

$$A = \{a, b\}$$

$$B = \{a, b, c, d\}$$

Then

$$A \subset B$$

because every element of A is in B and both sets are different.

1.8 Difference between Subset and Proper Subset

Feature	Subset ($\subseteq$)	Proper Subset ($\subset$)
Elements of first set are present in second set	Yes	Yes
Both sets may be equal	Yes	No
Extra elements in second set required	No	Yes
Example	$\{1, 2\} \subseteq \{1, 2\}$	$\{1, 2\} \subset \{1, 2, 3\}$

1.9 Important Facts

Every set is always a subset of itself.

$$A \subseteq A$$

However, a set can never be a proper subset of itself.

$$A \not\subset A$$

The empty set is a subset of every set.

$$\emptyset \subseteq A$$

Common Mistake

Students often think that

$$A \subseteq A$$

is false because both sets are the same. This is incorrect. A set is always a subset of itself. However,

$$A \subset A$$

is always false because a proper subset must be strictly smaller than the original set.

Example 1.5. Consider the following sets:

$$A = \{1, 2\}, \quad B = \{1, 2\}, \quad C = \{1, 2, 3\}$$

Observe carefully:

- A and B contain exactly the same elements.
- C contains all elements of A along with one additional element (3).

Now answer the following questions.

1. Is every element of A present in B ? **Yes.**

Therefore,

$$A \subseteq B$$

is True.

2. Is A smaller than B ? **No.**

Therefore,

$$A \subset B$$

is False.

3. Is every element of A present in C ? **Yes.**

Therefore,

$$A \subseteq C$$

is True.

4. Does C contain at least one extra element? **Yes.**

Therefore,

$$A \subset C$$

is also True.

Quick Summary

Remember the following shortcut:

- $\subseteq$ means **may be equal**.
- $\subset$ means **must be smaller**.

This single idea helps solve most examination questions on subsets.

Proof. The statement $\emptyset \subseteq A$ is true because there is no element of $\emptyset$ that can violate the subset condition. Also, every element of A is trivially an element of A , so $A \subseteq A$. $\square$

1.10 Cardinality

Definition 1.6. The cardinality of a finite set A , denoted by $|A|$ or $n(A)$, is the number of distinct elements in A .

Example 1.6. If

$$A = \{2, 4, 4, 6, 8\},$$

then $A = \{2, 4, 6, 8\}$ and therefore $|A| = 4$.

1.11 Power Set

Definition 1.7. The power set of A , denoted by $\mathcal{P}(A)$, is the set of all subsets of A :

$$\mathcal{P}(A) = \{X \mid X \subseteq A\}.$$

Example 1.7. For $A = \{a, b\}$,

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}.$$

Theorem 1.1 (Cardinality of a Power Set). *If $|A| = n$, then*

$$|\mathcal{P}(A)| = 2^n.$$

Proof. For each of the n elements, there are two independent choices: include it in a subset or exclude it. Therefore, the number of possible subsets is

$$\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}} = 2^n.$$

□

Example 1.8. If $|A| = 5$, then A has $2^5 = 32$ subsets. The number of proper subsets is $2^5 - 1 = 31$.

Exam Tip

For a set with n elements:

$$\begin{aligned} \text{number of subsets} &= 2^n, \\ \text{number of proper subsets} &= 2^n - 1, \\ \text{number of non-empty subsets} &= 2^n - 1, \\ \text{number of non-empty proper subsets} &= 2^n - 2. \end{aligned}$$

Quick Summary

A set is a well-defined collection of distinct objects. Sets may be represented using roster, set-builder, or descriptive form. Subset relations compare membership, cardinality counts distinct elements, and a set of cardinality n has exactly 2^n subsets.

2 Set Operations and Algebra of Sets

Learning Outcomes

After completing this section, students should be able to perform union, intersection, difference, complement and symmetric difference; verify set identities; use Venn diagrams; and apply the algebra of sets in problem solving.

Let A and B be subsets of a universal set U .

2.1 Union

Definition 2.1. The union of A and B is the set of elements belonging to at least one of the two sets:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

Here “or” is inclusive.

Example 2.1. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$, then

$$A \cup B = \{1, 2, 3, 4, 5, 6\}.$$

2.2 Intersection

Definition 2.2. The intersection of A and B is the set of elements common to both:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

For the preceding sets,

$$A \cap B = \{3, 4\}.$$

Two sets are **disjoint** if

$$A \cap B = \emptyset.$$

2.3 Difference

Definition 2.3. The difference $A - B$ contains elements that belong to A but not to B :

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}.$$

For $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$,

$$A - B = \{1, 2\}, \quad B - A = \{5, 6\}.$$

In general, set difference is not commutative.

2.4 Complement

Definition 2.4. The complement of A relative to U is

$$A^c = U - A = \{x \in U \mid x \notin A\}.$$

Example 2.2. If $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{1, 3, 5\}$, then

$$A^c = \{2, 4, 6\}.$$

2.5 Symmetric Difference

Definition 2.5. The symmetric difference of A and B contains elements belonging to exactly one of the sets:

$$A \Delta B = (A - B) \cup (B - A).$$

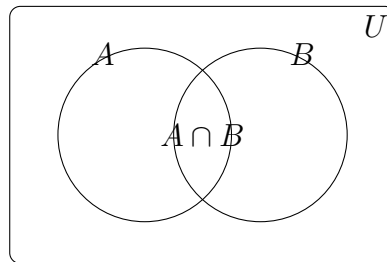
Equivalently,

$$A \Delta B = (A \cup B) - (A \cap B).$$

Example 2.3. For $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$,

$$A \Delta B = \{1, 2, 5, 6\}.$$

2.6 Venn Diagram Interpretation



A Venn diagram is useful for visual understanding, but algebraic proof is more reliable when several sets are involved.

2.7 Fundamental Laws of Set Algebra

For arbitrary subsets $A, B, C \subseteq U$:

Law	Identities
Commutative	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associative	$(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C)$
Distributive	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity	$A \cup \emptyset = A, \quad A \cap U = A$
Domination	$A \cup U = U, \quad A \cap \emptyset = \emptyset$
Idempotent	$A \cup A = A, \quad A \cap A = A$
Complement	$A \cup A^c = U, \quad A \cap A^c = \emptyset$
Double complement	$(A^c)^c = A$
Absorption	$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$
De Morgan	$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c$

2.8 Proof of a Set Identity by Element Method

Example 2.4. Prove

$$A - (B \cup C) = (A - B) \cap (A - C).$$

Proof. For an arbitrary element x ,

$$\begin{aligned}
 x \in A - (B \cup C) &\iff x \in A \text{ and } x \notin B \cup C \\
 &\iff x \in A \text{ and } (x \notin B \text{ and } x \notin C) \\
 &\iff (x \in A \text{ and } x \notin B) \text{ and } (x \in A \text{ and } x \notin C) \\
 &\iff x \in (A - B) \cap (A - C).
 \end{aligned}$$

Therefore,

$$A - (B \cup C) = (A - B) \cap (A - C).$$

2.9 Duality Principle

The dual of a set identity is obtained by interchanging

$$\cup \leftrightarrow \cap \quad \text{and} \quad U \leftrightarrow \emptyset.$$

For example, the dual of $A \cup \emptyset = A$ is $A \cap U = A$.

2.10 Computer Science Interpretation

Set operations appear naturally in data processing:

- union combines search results;
- intersection finds common users, records, or permissions;
- difference removes blocked or invalid items;
- symmetric difference detects changes between two versions;
- complement represents negation relative to a fixed universe.

Exam Tip

To prove $A = B$, show both $A \subseteq B$ and $B \subseteq A$, or use the element method:

$$x \in A \iff x \in B.$$

Do not rely only on a roughly drawn Venn diagram for a formal proof.

2.11 Solved Problems

Example 2.5. Let

$$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, \quad A = \{1, 2, 3, 4, 5\}, \quad B = \{4, 5, 6, 7\}.$$

Find $A \cup B$, $A \cap B$, $A - B$, $B - A$, A^c , and $A \triangle B$.

Solution.

$$A \cup B = \{1, 2, 3, 4, 5, 6, 7\},$$

$$A \cap B = \{4, 5\},$$

$$A - B = \{1, 2, 3\},$$

$$B - A = \{6, 7\},$$

$$A^c = \{6, 7, 8, 9, 10\},$$

$$A \triangle B = \{1, 2, 3, 6, 7\}.$$

Exercise 2.1. For $U = \{1, 2, \dots, 12\}$, $A = \{2, 4, 6, 8, 10, 12\}$, and $B = \{3, 6, 9, 12\}$, determine:

- (a) $A \cup B$,
- (b) $A \cap B$,
- (c) $A - B$,
- (d) $(A \cup B)^c$,
- (e) $A \Delta B$.

Quick Summary

Union corresponds to inclusive “or,” intersection to “and,” and complement to “not.” Difference is directional, while symmetric difference keeps elements occurring in exactly one set. Set identities can be proved rigorously using membership logic.

3 Cartesian Product

Learning Outcomes

After completing this section, students should be able to form ordered pairs and Cartesian products, determine their cardinalities, distinguish $A \times B$ from $B \times A$, and connect Cartesian products with relations, matrices, databases and coordinate systems.

3.1 Ordered Pairs

An ordered pair is written as (a, b) . Its order matters.

$$(a, b) = (c, d) \iff a = c \text{ and } b = d.$$

In general,

$$(a, b) \neq (b, a)$$

unless $a = b$.

Definition 3.1 (Cartesian Product). For sets A and B , the Cartesian product $A \times B$ is

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Example 3.1. If $A = \{1, 2\}$ and $B = \{x, y, z\}$, then

$$A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}.$$

Similarly,

$$B \times A = \{(x, 1), (x, 2), (y, 1), (y, 2), (z, 1), (z, 2)\}.$$

Thus $A \times B \neq B \times A$.

3.2 Cardinality of a Cartesian Product

Theorem 3.1. If A and B are finite sets, then

$$|A \times B| = |A| |B|.$$

Proof. For each of the $|A|$ possible first components, there are $|B|$ choices for the second component. By the multiplication principle, the total number of ordered pairs is $|A||B|$. $\square$

Example 3.2. If $|A| = 4$ and $|B| = 7$, then

$$|A \times B| = 4 \times 7 = 28.$$

3.3 Cartesian Powers

$$A^2 = A \times A,$$

and more generally,

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}}.$$

An element of A^n is an ordered n -tuple.

Example 3.3. If $B = \{0, 1\}$, then

$$B^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}.$$

Therefore $|B^3| = 2^3 = 8$.

3.4 Properties of Cartesian Products

For arbitrary sets A, B, C :

$$\begin{aligned} A \times (B \cup C) &= (A \times B) \cup (A \times C), \\ A \times (B \cap C) &= (A \times B) \cap (A \times C), \\ A \times (B - C) &= (A \times B) - (A \times C), \\ A \times \emptyset &= \emptyset, \\ \emptyset \times A &= \emptyset. \end{aligned}$$

Cartesian product is generally neither commutative nor associative in the strict ordered-pair sense.

3.5 Matrix View

If $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2\}$, the elements of $A \times B$ can be arranged in a rectangular grid:

	b_1	b_2
a_1	(a_1, b_1)	(a_1, b_2)
a_2	(a_2, b_1)	(a_2, b_2)
a_3	(a_3, b_1)	(a_3, b_2)

A binary relation from A to B is obtained by selecting some cells from this grid.

3.6 Applications in Computer Science

- A database relation consists of tuples drawn from Cartesian products of attribute domains.
- Pixel coordinates in an image belong to $X \times Y$.
- State-input pairs in an automaton belong to $Q \times \Sigma$.
- Edges in a directed graph form a subset of $V \times V$.
- Test cases for two independent input variables may be generated from $A \times B$.

Common Mistake

Do not treat (a, b) as the same object as $\{a, b\}$. An ordered pair preserves position; an ordinary set does not.

Exercise 3.1. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$.

- Write $A \times B$ and $B \times A$.
- Determine $|A \times B|$ and $|B \times A|$.
- Find A^2 .

Quick Summary

The Cartesian product $A \times B$ contains ordered pairs whose first component comes from A and second component from B . For finite sets, its cardinality is the product of cardinalities. Cartesian products provide the underlying universe for relations and tuples.

4 Inclusion–Exclusion Principle

Learning Outcomes

After completing this section, students should be able to count elements in overlapping sets, derive the two-set and three-set inclusion–exclusion formulas, solve survey and divisibility problems, and identify common counting errors.

4.1 Why Direct Addition Fails

If two finite sets overlap, adding their cardinalities counts every common element twice. Therefore, the overlap must be subtracted once.

Theorem 4.1 (Inclusion–Exclusion for Two Sets). *For finite sets A and B ,*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Proof. The quantity $|A| + |B|$ counts elements in $A \cap B$ twice: once as members of A and once as members of B . Subtracting $|A \cap B|$ corrects this double counting. $\square$

Example 4.1. In a class, 38 students study C++, 27 study Python, and 15 study both. The number studying at least one language is

$$38 + 27 - 15 = 50.$$

4.2 Exactly One and Neither

For two sets:

$$\begin{aligned} |A \text{ only}| &= |A| - |A \cap B|, \\ |B \text{ only}| &= |B| - |A \cap B|, \\ |\text{exactly one}| &= |A| + |B| - 2|A \cap B|. \end{aligned}$$

If the universal set has cardinality N , then

$$|\text{neither } A \text{ nor } B| = N - |A \cup B|.$$

Example 4.2. Among 100 students, 60 study Java, 45 study Python, and 25 study both.

$$|J \cup P| = 60 + 45 - 25 = 80.$$

Hence 20 study neither. Java only is $60 - 25 = 35$, Python only is $45 - 25 = 20$, and exactly one is $35 + 20 = 55$.

4.3 Three-Set Formula

Theorem 4.2 (Inclusion–Exclusion for Three Sets). *For finite sets A, B, C ,*

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| \\ &\quad + |A \cap B \cap C|. \end{aligned}$$

Why the Triple Intersection Is Added Back

After adding the three individual cardinalities, an element in all three sets is counted three times. Subtracting the three pairwise intersections removes it three times, leaving a count of zero. It must therefore be added once through $|A \cap B \cap C|$.

Example 4.3. In a survey of 120 students:

$$|C| = 65, \quad |J| = 55, \quad |P| = 50,$$

$$|C \cap J| = 25, \quad |C \cap P| = 20, \quad |J \cap P| = 18,$$

and $|C \cap J \cap P| = 10$. Then

$$\begin{aligned} |C \cup J \cup P| &= 65 + 55 + 50 - 25 - 20 - 18 + 10 \\ &= 117. \end{aligned}$$

Therefore, $120 - 117 = 3$ students study none of the three languages.

4.4 Finding Region-Wise Counts

Suppose $|A \cap B \cap C| = t$. Then:

$$|A \cap B \text{ only}| = |A \cap B| - t,$$

$$|A \cap C \text{ only}| = |A \cap C| - t,$$

$$|B \cap C \text{ only}| = |B \cap C| - t.$$

After these regions are known, the “only A ,” “only B ,” and “only C ” regions can be found by subtraction.

4.5 Divisibility Problem

Example 4.4. How many integers from 1 to 100 are divisible by 2 or 3?

Let

$$A = \{n \leq 100 \mid 2 \mid n\}, \quad B = \{n \leq 100 \mid 3 \mid n\}.$$

Then

$$|A| = \left\lfloor \frac{100}{2} \right\rfloor = 50, \quad |B| = \left\lfloor \frac{100}{3} \right\rfloor = 33.$$

Numbers divisible by both 2 and 3 are divisible by $\text{lcm}(2, 3) = 6$:

$$|A \cap B| = \left\lfloor \frac{100}{6} \right\rfloor = 16.$$

Therefore,

$$|A \cup B| = 50 + 33 - 16 = 67.$$

Example 4.5. How many integers from 1 to 200 are divisible by 4, 6, or 10?

Let A, B, C denote divisibility by 4, 6, and 10, respectively. Then

$$|A| = 50, \quad |B| = 33, \quad |C| = 20.$$

Pairwise intersections are determined by least common multiples:

$$|A \cap B| = \left\lfloor \frac{200}{12} \right\rfloor = 16,$$

$$|A \cap C| = \left\lfloor \frac{200}{20} \right\rfloor = 10,$$

$$|B \cap C| = \left\lfloor \frac{200}{30} \right\rfloor = 6.$$

The triple intersection uses $\text{lcm}(4, 6, 10) = 60$:

$$|A \cap B \cap C| = \left\lfloor \frac{200}{60} \right\rfloor = 3.$$

Hence,

$$50 + 33 + 20 - 16 - 10 - 6 + 3 = 74.$$

Common Mistake

Pairwise intersection counts usually include members of the triple intersection unless the question explicitly says “only.” Do not subtract the triple intersection before checking the wording.

Exercise 4.1. In a group of 150 students, 85 know C, 70 know Java, and 55 know Python. Further, 35 know both C and Java, 30 know both C and Python, 25 know both Java and Python, and 15 know all three. Find:

- (a) the number who know at least one language;
- (b) the number who know exactly one language;
- (c) the number who know none.

Quick Summary

The inclusion–exclusion principle corrects overcounting caused by overlaps. For two sets, subtract the common part once. For three sets, subtract pairwise intersections and add the triple intersection back. In divisibility problems, intersections are found using least common multiples.

5 Fuzzy Sets

Learning Outcomes

After studying this section, students will be able to:

- Understand why fuzzy sets were introduced.
- Differentiate between crisp sets and fuzzy sets.
- Represent a fuzzy set using membership values.
- Perform basic operations on fuzzy sets.
- Identify practical applications of fuzzy sets.

5.1 Introduction

In the previous sections, we studied ordinary sets (also called **crisp sets**). In a crisp set, an element either belongs to the set or it does not belong to the set.

For example,

$$A = \{\text{Students having age 18 years or above}\}$$

If Rahul is 19 years old, then Rahul belongs to the set. If Mohit is 17 years old, then Mohit does not belong to the set.

There are only two possibilities:

- Belongs to the set
- Does not belong to the set

There is nothing in between.

5.2 Need for Fuzzy Sets

Although crisp sets work well for many situations, they fail to represent concepts that do not have clear boundaries.

Consider the following questions.

- Who is a tall student?
- Which day is hot?
- Which mobile phone is expensive?
- Which employee is hardworking?
- Which movie is good?

Different people may give different answers. For example, one person may consider a student of height 170 cm as tall, while another person may say that only students taller than 180 cm are tall. Hence, the answer is not simply **Yes** or **No**. Such situations require a new type of set called a **Fuzzy Set**.

Real-Life Analogy

Suppose a teacher asks,

“Who are the intelligent students in this class?”

Can we answer this question with only “Yes” or “No”?

No.

Some students may be highly intelligent, some may be moderately intelligent, while others may still be learning.

Therefore, it is more meaningful to express the degree of intelligence rather than making a strict decision.

Fuzzy sets are designed to handle such situations.

5.3 What is a Fuzzy Set?

A fuzzy set is a set in which every element is assigned a **degree of membership**.

Instead of saying

Belongs to the set

OR

Does not belong to the set

we specify

How much does it belong?

The degree of membership is represented by a number between 0 and 1.

Membership Value	Meaning
0	Does not belong
0.2	Very small membership
0.5	Partially belongs
0.8	Mostly belongs
1	Completely belongs

Definition 5.1. A **Fuzzy Set** is a set in which every element is associated with a membership value between 0 and 1.

The membership value indicates the degree to which an element belongs to the set.

5.4 Example of a Fuzzy Set

Suppose we want to describe the set of “Tall Students”.

Student	Height (cm)	Membership in Tall Students
A	150	0.0
B	160	0.2
C	170	0.5
D	180	0.8
E	190	1.0

This fuzzy set can be written as

$$T = \{0.0/150, 0.2/160, 0.5/170, 0.8/180, 1.0/190\}$$

Here,

- 150 cm is not considered tall.
- 170 cm is moderately tall.
- 190 cm is completely tall.

Notice that the numbers before the slash represent the degree of membership. The slash does **not** represent division.

5.5 Crisp Set vs Fuzzy Set

Feature	Crisp Set	Fuzzy Set
Membership	Yes or No	Degree of Membership
Values Allowed	0 or 1	Any value between 0 and 1
Boundary	Sharp	Gradual
Suitable For	Exact situations	Uncertain situations
Example	Prime Numbers	Tall Students

5.6 Basic Operations on Fuzzy Sets

Operations on fuzzy sets are similar to ordinary sets. The only difference is that we use membership values.

5.6.1 Union

The union of two fuzzy sets is obtained by taking the **larger membership value**.

$$A \cup B = \text{Maximum Membership}$$

Example

Student	Smart	Hardworking	Smart OR Hardworking
A	0.7	0.5	0.7
B	0.4	0.8	0.8
C	0.6	0.6	0.6

5.6.2 Intersection

The intersection of two fuzzy sets is obtained by taking the **smaller membership value**.

$$A \cap B = \text{Minimum Membership}$$

Using the previous table,

Student	Smart	Hardworking	Smart AND Hardworking
A	0.7	0.5	0.5
B	0.4	0.8	0.4
C	0.6	0.6	0.6

5.6.3 Complement

The complement represents “NOT”. It is calculated as

$$1 - \text{Membership Value}$$

Example

Student	Smart	Not Smart
A	0.8	0.2
B	0.6	0.4
C	0.3	0.7

5.7 Applications of Fuzzy Sets

Fuzzy sets are widely used in computer science and engineering. Some common applications are:

- Automatic washing machines
- Air conditioners
- Camera autofocus systems
- Medical diagnosis
- Artificial Intelligence
- Robotics
- Decision Support Systems
- Pattern Recognition
- Image Processing
- Risk Analysis

5.8 Solved Example

Example 5.1. Consider the fuzzy set

$$A = \{0.2/a, 0.5/b, 0.9/c\}$$

Find the complement of the fuzzy set.

Solution Subtract every membership value from 1.

$$A' = \{0.8/a, 0.5/b, 0.1/c\}$$

Hence, the complement is

$$\boxed{\{0.8/a, 0.5/b, 0.1/c\}}$$

5.9 Key Points to Remember

Quick Summary

- Ordinary sets allow only two membership values: 0 or 1.
- Fuzzy sets allow any membership value between 0 and 1.
- Fuzzy sets are useful for representing uncertain or vague concepts.
- Union is obtained by taking the maximum value.
- Intersection is obtained by taking the minimum value.
- Complement is obtained by subtracting the membership value from 1.

5.10 Practice Questions

1. Define a fuzzy set.
2. Differentiate between a crisp set and a fuzzy set.
3. Explain the need for fuzzy sets.
4. Find the complement of

$$A = \{0.3/a, 0.6/b, 0.9/c\}$$

5. State any four applications of fuzzy sets.

6 Multisets

Learning Outcomes

After studying this section, students will be able to:

- Understand why multisets were introduced.
- Differentiate between a set and a multiset.
- Represent multisets.
- Perform basic multiset operations.
- Explain applications of multisets in Computer Science.

6.1 Introduction

In the previous sections, we studied ordinary sets. One important property of a set is that it does **not allow duplicate elements**.

For example,

$$A = \{1, 2, 3, 4\}$$

Now suppose we write

$$A = \{1, 1, 2, 2, 3, 4, 4\}$$

Even though some elements are repeated, the set is still written as

$$A = \{1, 2, 3, 4\}$$

because repeated elements are ignored.

6.2 Why Do We Need Multisets?

Consider the following situations.

1. A shopping basket contains

$$\{\text{Apple, Apple, Apple, Banana, Orange, Orange}\}$$

Should we ignore the repeated apples? Obviously not.

2. A library contains five copies of the same book.
3. A supermarket has 100 packets of milk.
4. A document contains the word **Computer** fifty times.

In all these cases, repeated elements are important. Hence, ordinary sets cannot represent such data properly.

Remember

A **Multiset** is a collection in which duplicate elements are allowed. Unlike an ordinary set, a multiset keeps track of how many times each element appears.

6.3 Definition

Definition 6.1. A multiset is a collection of elements in which one or more elements may occur multiple times. The number of times an element occurs is called its **multiplicity** or **frequency**.

6.4 Representation of a Multiset

Consider the shopping basket

$$\{\text{Apple, Apple, Apple, Banana, Orange, Orange}\}$$

Instead of writing repeated elements again and again, we can write

$$M = \{3/\text{Apple}, 1/\text{Banana}, 2/\text{Orange}\}$$

Here,

- Apple appears three times.
- Banana appears one time.
- Orange appears two times.

The number before the slash indicates the number of occurrences.

6.5 Another Example

Suppose the marks obtained by six students are

$$\{80, 75, 80, 90, 75, 80\}$$

The corresponding multiset is

$$M = \{3/80, 2/75, 1/90\}$$

which shows that

- 80 occurs three times.
- 75 occurs two times.
- 90 occurs only once.

6.6 Difference between Set and Multiset

Feature	Set	Multiset
Duplicate elements	Not Allowed	Allowed
Repeated values	Ignored	Counted
Stores frequency	No	Yes
Example	$\{1, 2, 3\}$	$\{1, 1, 2, 2, 3\}$

6.7 Basic Operations on Multisets

Suppose

$$M = \{2/a, 1/b, 3/c\}$$

and

$$N = \{1/a, 3/b, 2/c\}$$

Union

For union, we keep the larger occurrence of every element.

$$M \cup N = \{2/a, 3/b, 3/c\}$$

Intersection

For intersection, we keep the smaller occurrence.

$$M \cap N = \{1/a, 1/b, 2/c\}$$

Addition

In addition, the occurrences are added.

$$M \uplus N = \{3/a, 4/b, 5/c\}$$

Difference

Subtract the occurrences. If the result becomes negative, replace it with zero.

$$M - N = \{1/a, 0/b, 1/c\}$$

Usually, elements having zero occurrences are omitted. Hence,

$$M - N = \{1/a, 1/c\}$$

6.8 Applications of Multisets

Multisets are widely used in Computer Science. Some applications are

- Shopping carts
- Inventory management
- Database systems
- Search engines
- Word frequency analysis
- Text mining
- Compiler design
- Data analytics
- Library management systems

6.9 Solved Example

Example 6.1. Let

$$M = \{2/a, 1/b, 3/c\}$$

and

$$N = \{1/a, 2/b, 1/c\}$$

Find

$$M \cup N$$

and

$$M \cap N.$$

Solution For union, select the larger occurrence.

$$M \cup N = \{2/a, 2/b, 3/c\}$$

For intersection, select the smaller occurrence.

$$M \cap N = \{1/a, 1/b, 1/c\}$$

6.10 Multiset vs Fuzzy Set

Students often confuse these two concepts.

Multiset	Fuzzy Set
Number before slash indicates frequency	Number before slash indicates membership
$3/a$ means a occurs three times	$0.3/a$ means a belongs with degree 0.3
Duplicates are allowed	Duplicates are not considered

Quick Summary

- Ordinary sets do not allow duplicate elements.
- Multisets allow duplicate elements.
- The number of occurrences is called multiplicity.
- Union uses the maximum occurrence.
- Intersection uses the minimum occurrence.
- Multisets are widely used in databases, shopping carts and inventory systems.

6.11 Practice Questions

1. Define a multiset.
2. Why are multisets needed?
3. Differentiate between a set and a multiset.
4. Differentiate between a multiset and a fuzzy set.

5. Find the union and intersection of

$$M = \{2/a, 3/b, 1/c\}$$

and

$$N = \{1/a, 2/b, 4/c\}.$$

6. State any four applications of multisets.

7 Binary Relations and Their Representation

Learning Outcomes

After completing this section, students should be able to define a binary relation as a subset of a Cartesian product, determine domain and range, represent relations using ordered pairs, zero-one matrices and directed graphs, and convert between these representations.

7.1 Motivation

Sets describe collections; relations describe connections between elements. Examples include:

- one student follows another on a social network;
- one webpage links to another;
- one task depends on another;
- one number divides another;
- one database record is associated with another.

Definition 7.1 (Binary Relation). Let A and B be sets. A binary relation R from A to B is any subset of $A \times B$:

$$R \subseteq A \times B.$$

If $(a, b) \in R$, we write aRb and say that a is related to b .

A relation *on* a set A is a subset of $A \times A$.

Example 7.1. Let $A = \{1, 2, 3\}$ and $B = \{a, b\}$. Then

$$R = \{(1, a), (2, a), (2, b)\}$$

is a relation from A to B because every ordered pair belongs to $A \times B$.

7.2 Domain, Codomain and Range

For a relation $R \subseteq A \times B$:

$$\begin{aligned} \text{dom}(R) &= \{a \in A \mid \exists b \in B : (a, b) \in R\}, \\ \text{ran}(R) &= \{b \in B \mid \exists a \in A : (a, b) \in R\}. \end{aligned}$$

The set B is the codomain, while the range consists only of second components that actually occur.

Example 7.2. For

$$R = \{(1, a), (2, a), (2, b)\} \subseteq \{1, 2, 3\} \times \{a, b, c\},$$

we have

$$\text{dom}(R) = \{1, 2\}, \quad \text{ran}(R) = \{a, b\},$$

while the codomain is $\{a, b, c\}$.

7.3 Number of Relations

Theorem 7.1. *If $|A| = m$ and $|B| = n$, then the number of distinct relations from A to B is*

$$2^{mn}.$$

Proof. The Cartesian product $A \times B$ contains mn ordered pairs. A relation is any subset of this Cartesian product. Therefore, the number of relations is the number of subsets of an mn -element set, namely 2^{mn} . $\square$

Example 7.3. If $|A| = 3$ and $|B| = 2$, then there are

$$2^{3 \cdot 2} = 2^6 = 64$$

distinct relations from A to B .

7.4 Commonly Defined Relations

7.4.1 Relation Defined by a Formula

Let $A = \{1, 2, 3, 4\}$ and define

$$aRb \iff a < b.$$

Then

$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}.$$

7.4.2 Divisibility Relation

On $A = \{1, 2, 3, 6\}$, define

$$aRb \iff a \mid b.$$

Then

$$R = \{(1, 1), (1, 2), (1, 3), (1, 6), (2, 2), (2, 6), (3, 3), (3, 6), (6, 6)\}.$$

7.5 Representation by Ordered Pairs

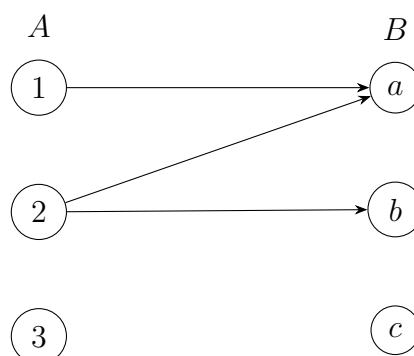
The most direct representation explicitly lists every pair in the relation:

$$R = \{(1, 1), (1, 2), (2, 3)\}.$$

This is convenient for small relations but becomes lengthy for large sets.

7.6 Arrow Diagram

For a relation from A to B , write the elements of A and B in two columns and draw an arrow from a to b whenever $(a, b) \in R$.



The arrow diagram above represents

$$R = \{(1, a), (2, a), (2, b)\}.$$

7.7 Zero–One Matrix Representation

Let

$$A = \{a_1, a_2, \dots, a_m\}, \quad B = \{b_1, b_2, \dots, b_n\}.$$

The matrix of $R \subseteq A \times B$ is the $m \times n$ matrix $M_R = [m_{ij}]$ defined by

$$m_{ij} = \begin{cases} 1, & (a_i, b_j) \in R, \\ 0, & (a_i, b_j) \notin R. \end{cases}$$

Example 7.4. For

$$A = \{1, 2, 3\}, \quad B = \{a, b, c\}, \quad R = \{(1, a), (2, a), (2, b), (3, c)\},$$

using the displayed orders of A and B ,

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Rows correspond to first components and columns to second components.

Common Mistake

A relation matrix depends on the chosen order of elements. Always write the row and column order before constructing or interpreting the matrix.

7.8 Directed Graph Representation

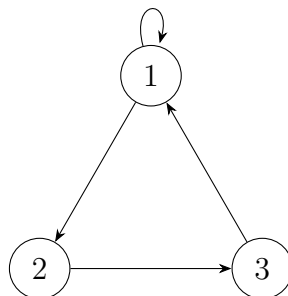
If R is a relation on A , it can be represented by a directed graph (digraph):

- each element of A is a vertex;
- draw an arrow $a \rightarrow b$ whenever $(a, b) \in R$;
- a pair (a, a) is represented by a loop at a .

Example 7.5. Let

$$A = \{1, 2, 3\}, \quad R = \{(1, 1), (1, 2), (2, 3), (3, 1)\}.$$

Its digraph is:



7.9 Converting Between Representations

To convert ordered pairs to a matrix, place 1 in each corresponding row-column position. To convert a matrix to a digraph, create one vertex per row/column element and draw an edge for every entry equal to 1.

Example 7.6. Let

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

for the ordered set $A = \{a, b, c\}$. Then

$$R = \{(a, a), (a, b), (b, b), (b, c), (c, a)\}.$$

7.10 Special Relations Introduced at the Representation Stage

For a set A :

- the **empty relation** is $\emptyset$;
- the **universal relation** is $A \times A$;
- the **identity relation** is

$$I_A = \{(a, a) \mid a \in A\}.$$

Their matrices are respectively the zero matrix, the all-one matrix, and the identity matrix.

7.11 Relation versus Function

Every function from A to B is a relation from A to B , but not every relation is a function. A function must associate each element of A with exactly one element of B .

Example 7.7.

$$R = \{(1, a), (2, b), (3, b)\}$$

is a function from $\{1, 2, 3\}$ to $\{a, b\}$. However,

$$S = \{(1, a), (1, b), (2, a)\}$$

is not a function because 1 has two images and 3 has no image if the domain is $\{1, 2, 3\}$.

7.12 Computer Science Applications

- adjacency matrices represent graph connectivity;
- database tables represent relations among attribute values;
- access-control matrices relate users to permitted resources;
- dependency graphs relate modules, packages or tasks;
- social networks relate users through following, friendship or messaging links;
- Warshall's algorithm later computes reachability from a relation matrix.

Exam Tip

When a relation is defined by a condition such as $aRb \iff a + b$ is even, test every ordered pair in $A \times A$ systematically. A table or matrix prevents omissions.

7.13 Solved Problems

Example 7.8. Let $A = \{1, 2, 3, 4\}$ and define aRb if $a + b$ is even. List R and write its matrix.

Solution. A sum is even when both numbers have the same parity. Therefore,

$$R = \{(1, 1), (1, 3), (2, 2), (2, 4), \\ (3, 1), (3, 3), (4, 2), (4, 4)\}.$$

Using the order 1, 2, 3, 4,

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

Example 7.9. Let $A = \{1, 2, 3, 6\}$ and R be the divisibility relation. Determine the domain and range.

Solution. Every element divides itself, so every element appears as a first and second component. Hence,

$$\text{dom}(R) = \text{ran}(R) = A.$$

Exercise 7.1. Let $A = \{1, 2, 3, 4\}$ and

$$R = \{(a, b) \in A \times A \mid a \leq b\}.$$

- (a) List all ordered pairs of R .
- (b) Write the zero-one matrix of R .
- (c) Draw its directed graph.
- (d) Find its domain and range.

Quick Summary

A binary relation from A to B is any subset of $A \times B$. It may be represented by ordered pairs, an arrow diagram, a zero-one matrix, or a directed graph. Domain and range describe the first and second components that actually occur. These representations form the basis for studying relation properties, composition, closure and reachability.

8 Revision Exercises for the Completed Topics

8.1 Objective Questions

1. If $|A| = 6$, then $|\mathcal{P}(A)|$ is:
 - (a) 12
 - (b) 36
 - (c) 64
 - (d) 720
2. If $A \subseteq B$, then $A \cap B$ equals:
 - (a) A
 - (b) B
 - (c) $\emptyset$
 - (d) A^c
3. Which operation is not commutative in general?
 - (a) Union
 - (b) Intersection
 - (c) Difference
 - (d) Symmetric difference
4. If $|A| = 3$ and $|B| = 5$, then the number of relations from A to B is:
 - (a) 15
 - (b) 2^8
 - (c) 2^{15}
 - (d) 3^5
5. The matrix of the identity relation on a four-element set is:
 - (a) a zero matrix
 - (b) an all-one matrix
 - (c) a diagonal identity matrix
 - (d) a triangular matrix

8.2 Short-Answer Questions

1. Distinguish equal sets from equivalent sets with an example.
2. Prove that $A - B = A \cap B^c$.
3. State both of De Morgan's laws for sets.
4. Explain why $A \times B$ need not equal $B \times A$.
5. Define a binary relation from A to B .

6. Distinguish the range of a relation from its codomain.
7. Explain the relation between a digraph and a zero-one relation matrix.
8. How many relations are possible on a four-element set?

8.3 Long-Answer and Numerical Questions

1. Let

$$U = \{1, 2, \dots, 20\}, \quad A = \{x \in U \mid 2 \mid x\}, \quad B = \{x \in U \mid 3 \mid x\}.$$

Find $A \cup B$, $A \cap B$, $A - B$, $B - A$, and $(A \cup B)^c$.

2. Prove the distributive law

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

by the element method.

3. In a survey of 200 students, 120 use Windows, 90 use Linux, and 50 use both. Find the number using at least one operating system, exactly one, and neither.
4. Among 300 users, 180 use application A , 150 use application B , and 120 use application C . Further, 80 use both A and B , 65 use both A and C , 60 use both B and C , and 30 use all three. Determine the number using at least one and none.
5. Let $A = \{1, 2, 3, 4\}$ and define R by aRb if a divides b . List R , write its matrix, and draw its digraph.
6. Let

$$M_R = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}.$$

For the ordered set $A = \{a, b, c, d\}$, list all ordered pairs in R and draw the corresponding digraph.

Exam Tip

Answer key for the objective questions: 1(c), 2(a), 3(c), 4(c), 5(c).

9 Properties and Types of Relations

Learning Outcomes

Students should be able to test a binary relation for reflexivity, irreflexivity, symmetry, asymmetry, antisymmetry and transitivity, and identify common special relations.

Let $R \subseteq A \times A$ be a binary relation on A .

9.1 Reflexive and Irreflexive Relations

Definition 9.1. R is **reflexive** if

$$\forall a \in A, \quad (a, a) \in R.$$

R is **irreflexive** if

$$\forall a \in A, \quad (a, a) \notin R.$$

Examples: $\leq$ is reflexive; $<$ is irreflexive.

9.2 Symmetric, Asymmetric and Antisymmetric Relations

Definition 9.2. R is **symmetric** if

$$(a, b) \in R \Rightarrow (b, a) \in R.$$

It is **asymmetric** if

$$(a, b) \in R \Rightarrow (b, a) \notin R.$$

It is **antisymmetric** if

$$(a, b) \in R \wedge (b, a) \in R \Rightarrow a = b.$$

Critical Difference

Antisymmetric does not mean “not symmetric”. A relation may be both symmetric and antisymmetric, for example the identity relation. Antisymmetry only prohibits two-way relatedness between distinct elements.

9.3 Transitive Relations

Definition 9.3. R is **transitive** if

$$(a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R.$$

9.4 Testing from a Matrix

For a relation matrix M_R :

- reflexive: every diagonal entry is 1,
- irreflexive: every diagonal entry is 0,
- symmetric: $M_R = M_R^T$,
- antisymmetric: for $i \neq j$, not both m_{ij} and m_{ji} are 1,
- transitive: Boolean product $M_R \odot M_R \leq M_R$.

9.5 Testing from a Digraph

- reflexive: loop at every vertex,
- symmetric: every arrow has a reverse arrow,
- antisymmetric: no two distinct vertices have arrows in both directions,
- transitive: every directed path of length two has a direct shortcut edge.

9.6 Common Types of Relations

Empty relation $\emptyset$.

Universal relation

$$A \times A.$$

Identity relation

$$I_A = \{(a, a) \mid a \in A\}.$$

Inverse relation $R^{-1} = \{(b, a) \mid (a, b) \in R\}.$

Complement relation

$$R^c = (A \times A) - R.$$

Equivalence relation

reflexive, symmetric and transitive.

Partial order relation

reflexive, antisymmetric and transitive.

9.7 Property Comparison Table

Property	Formal test	Typical example
Reflexive	aRa for every a	$\leq$, equality
Irreflexive	aRa for no a	$<$, proper subset
Symmetric	$aRb \Rightarrow bRa$	same age, sibling relation
Asymmetric	$aRb \Rightarrow \neg(bRa)$	$<$, parent of
Antisymmetric	$aRb \wedge bRa \Rightarrow a = b$	$\leq$, divisibility
Transitive	$aRb \wedge bRc \Rightarrow aRc$	$\leq$, divisibility

9.8 Solved Classification Problems

Example 9.1. On $A = \{1, 2, 3\}$, let

$$R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}.$$

R is reflexive, symmetric and transitive. It is not antisymmetric because $(1, 2)$ and $(2, 1)$ occur with $1 \neq 2$. Hence R is an equivalence relation.

Example 9.2. On positive integers, define aRb if a divides b . The relation is reflexive, antisymmetric and transitive, but not symmetric. Therefore it is a partial order.

Exam Tip

When testing transitivity on a finite relation, do not check random triples. For each pair (a, b) and each pair beginning with b , verify the required pair (a, c) .

10 Equality, Inverse and Composite Relations

10.1 Equality of Relations

Definition 10.1. Two relations R and S from A to B are equal if they contain exactly the same ordered pairs:

$$R = S \iff R \subseteq S \text{ and } S \subseteq R.$$

Equality can also be checked by identical matrices or identical directed edges.

10.2 Inverse Relation

If $R \subseteq A \times B$, then

$$R^{-1} = \{(b, a) \in B \times A \mid (a, b) \in R\}.$$

Important properties:

$$\begin{aligned} (R^{-1})^{-1} &= R, \\ (R \cup S)^{-1} &= R^{-1} \cup S^{-1}, \\ (R \cap S)^{-1} &= R^{-1} \cap S^{-1}. \end{aligned}$$

10.3 Composite Relations

Definition 10.2. Let $R \subseteq A \times B$ and $S \subseteq B \times C$. The composite relation $S \circ R$ is

$$S \circ R = \{(a, c) \in A \times C \mid \exists b \in B : (a, b) \in R \wedge (b, c) \in S\}.$$

The order matters: first apply R , then S .

Example 10.1. Let

$$R = \{(1, a), (2, b), (3, a)\}, \quad S = \{(a, x), (a, y), (b, z)\}.$$

Then

$$S \circ R = \{(1, x), (1, y), (2, z), (3, x), (3, y)\}.$$

10.4 Matrix Method for Composition

If M_R and M_S are relation matrices, then

$$M_{S \circ R} = M_R \odot M_S,$$

where $\odot$ denotes Boolean matrix multiplication:

$$c_{ij} = \bigvee_k (a_{ik} \wedge b_{kj}).$$

10.5 Powers of a Relation

For a relation R on A :

$$R^1 = R, \quad R^{n+1} = R^n \circ R.$$

R^n represents reachability by a path of exactly n relation steps.

Proposition 10.1. *Relation composition is associative:*

$$T \circ (S \circ R) = (T \circ S) \circ R.$$

It is not generally commutative.

11 Recursive Definition of Relations

Learning Outcomes

Students should understand how a relation can be generated from base pairs and recursive rules, and how recursive definitions describe infinitely many related pairs using finite rules.

11.1 General Form

A recursive definition normally has three parts:

1. **Basis step:** specifies initial pairs.
2. **Recursive step:** generates new pairs from existing pairs.
3. **Extremal clause:** states that no other pairs belong except those forced by the first two clauses.

Example 11.1 (The Relation $\leq$ on $\mathbb{N}$). Define R recursively by:

1. $(n, n) \in R$ for every $n \in \mathbb{N}$.
2. If $(a, b) \in R$, then $(a, b + 1) \in R$.
3. No other pairs are in R .

This generates exactly the relation $a \leq b$.

Example 11.2 (Ancestor Relation). Let P be the parent relation. Define ancestor relation A by:

1. If xPy , then xAy .
2. If xAy and yPz , then xAz .

The resulting relation is the transitive closure of the parent relation.

11.2 Recursive Relations in Computer Science

Recursive relation definitions occur in parse trees, graph reachability, file-system ancestry, program dependency, grammar derivations and inductively defined data structures.

12 Equivalence Relations and Partitions

Learning Outcomes

Students should be able to verify equivalence relations, compute equivalence classes, construct partitions, and explain the one-to-one correspondence between equivalence relations and partitions.

Definition 12.1. A relation R on A is an equivalence relation if it is reflexive, symmetric and transitive.

12.1 Equivalence Classes

For $a \in A$, the equivalence class of a is

$$[a] = \{x \in A \mid xRa\}.$$

Theorem 12.1. If R is an equivalence relation on A , then for any $a, b \in A$, either

$$[a] = [b]$$

or

$$[a] \cap [b] = \emptyset.$$

Proof. Suppose $[a] \cap [b] \neq \emptyset$. Let x belong to the intersection. Then xRa and xRb . By symmetry, aRx , and by transitivity, aRb . Now take any $y \in [a]$. Then yRa and aRb , so yRb . Thus $y \in [b]$, showing $[a] \subseteq [b]$. Similarly $[b] \subseteq [a]$. Hence $[a] = [b]$. $\square$

12.2 Congruence Modulo n

For integers, define

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

This is an equivalence relation. Its classes are

$$[0], [1], \dots, [n-1].$$

For modulo 3:

$$\begin{aligned} [0] &= \{\dots, -6, -3, 0, 3, 6, \dots\}, \\ [1] &= \{\dots, -5, -2, 1, 4, 7, \dots\}, \\ [2] &= \{\dots, -4, -1, 2, 5, 8, \dots\}. \end{aligned}$$

12.3 Partitions

Definition 12.2. A partition of a nonempty set A is a collection $\Pi = \{A_i \mid i \in I\}$ of nonempty subsets such that:

1. $A_i \cap A_j = \emptyset$ whenever $i \neq j$,
2. $\bigcup_{i \in I} A_i = A$.

Theorem 12.2 (Fundamental Correspondence). *Every equivalence relation on A determines a partition of A into equivalence classes. Conversely, every partition of A determines an equivalence relation by declaring two elements related exactly when they lie in the same block.*

Example 12.1. Let $A = \{1, 2, 3, 4, 5, 6\}$ and partition

$$\Pi = \{\{1, 3, 5\}, \{2, 4\}, \{6\}\}.$$

The corresponding equivalence relation contains all ordered pairs of elements from the same block.

12.4 Quotient Set

The set of all equivalence classes is denoted

$$A/R = \{[a] \mid a \in A\}$$

and is called the quotient set.

Common Mistake

Different representatives can define the same equivalence class. Do not count $[a]$ and $[b]$ separately when aRb .

13 Closure of Relations

Learning Outcomes

Students should be able to construct reflexive, symmetric and transitive closures and justify why each is the smallest relation containing the original relation with the required property.

Definition 13.1. The closure of a relation R with respect to a property P is the smallest relation S such that $R \subseteq S$ and S has property P .

13.1 Reflexive Closure

For a relation R on A ,

$$R_{\text{ref}} = R \cup I_A.$$

In matrix form, change every diagonal 0 to 1.

13.2 Symmetric Closure

$$R_{\text{sym}} = R \cup R^{-1}.$$

In a digraph, add the reverse of every directed edge.

13.3 Transitive Closure

The transitive closure of R , denoted R^+ , is

$$R^+ = R \cup R^2 \cup R^3 \cup \dots.$$

For a finite set with n elements, it is sufficient to use

$$R^+ = R \cup R^2 \cup \dots \cup R^n.$$

The reflexive transitive closure is

$$R^* = I_A \cup R^+.$$

Example 13.1. Let

$$R = \{(1, 2), (2, 3)\}$$

on $A = \{1, 2, 3\}$. Then

$$R_{\text{ref}} = R \cup \{(1, 1), (2, 2), (3, 3)\},$$

$$R_{\text{sym}} = R \cup \{(2, 1), (3, 2)\},$$

$$R^+ = R \cup \{(1, 3)\}.$$

13.4 Applications

Closures are used in graph reachability, database dependencies, prerequisite analysis, access-control inheritance, call graphs, package dependencies and compiler optimization.

14 Warshall's Algorithm

Learning Outcomes

Students should be able to execute Warshall's algorithm manually, produce intermediate matrices, prove its correctness informally, and state its time and space complexity.

14.1 Purpose

Warshall's algorithm computes the transitive closure of a directed graph or binary relation from its adjacency matrix.

Let $W^{(0)} = M_R$. For $k = 1, 2, \dots, n$,

$$w_{ij}^{(k)} = w_{ij}^{(k-1)} \vee (w_{ik}^{(k-1)} \wedge w_{kj}^{(k-1)}).$$

Here, $w_{ij}^{(k)} = 1$ means there is a path from i to j using only vertices $1, 2, \dots, k$ as intermediate vertices.

14.2 Algorithm

Algorithm 1 Warshall's Algorithm

Require: Adjacency matrix W of size $n \times n$

```

1: for  $k = 1$  to  $n$  do
2:   for  $i = 1$  to  $n$  do
3:     for  $j = 1$  to  $n$  do
4:        $W[i, j] \leftarrow W[i, j] \vee (W[i, k] \wedge W[k, j])$ 
5:     end for
6:   end for
7: end for
8: return  $W$ 

```

14.3 Detailed Example

Let

$$W^{(0)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Using vertex 1 as an intermediate gives

$$W^{(1)} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Using vertex 2:

$$W^{(2)} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Using vertex 3:

$$W^{(3)} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Using vertex 4 introduces no new reachability:

$$W^{(4)} = W^{(3)}.$$

Thus vertices 1, 2 and 3 are mutually reachable and can reach vertex 4.

14.4 Complexity

The algorithm uses three nested loops:

$$T(n) = \Theta(n^3).$$

In-place implementation uses $\Theta(n^2)$ space.

14.5 C++ Implementation

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    int n;
    cin >> n;
    vector<vector<int>> w(n, vector<int>(n));

    for (int i = 0; i < n; ++i)
        for (int j = 0; j < n; ++j)
            cin >> w[i][j];

    for (int k = 0; k < n; ++k)
        for (int i = 0; i < n; ++i)
            for (int j = 0; j < n; ++j)
                w[i][j] = w[i][j] || (w[i][k] && w[k][j]);

    for (const auto& row : w) {
        for (int value : row) cout << value << ' ';
        cout << '\n';
    }
}
```

Exam Tip

In handwritten solutions, clearly label $W^{(0)}, W^{(1)}, \dots, W^{(n)}$. Marks are often awarded for intermediate matrices, not only the final answer.

15 Partially Ordered Sets

Learning Outcomes

Students should be able to identify partial orders, compare partial, total and well orders, and determine minimal, maximal, least, greatest, upper-bound and lower-bound elements.

Definition 15.1. A relation $\preceq$ on a set P is a partial order if it is reflexive, antisymmetric and transitive. The pair $(P, \preceq)$ is called a partially ordered set or poset.

15.1 Standard Examples

1. $(\mathcal{P}(S), \subseteq)$.
2. $(\mathbb{N}, |)$ where $a \mid b$ means a divides b .
3. $(\mathbb{R}, \leq)$.
4. Strings ordered by the prefix relation.
5. Tasks ordered by prerequisite dependency.

15.2 Comparable and Incomparable Elements

Two elements $a, b \in P$ are comparable if $a \preceq b$ or $b \preceq a$. Otherwise they are incomparable.

Definition 15.2 (Total Order). A partial order is total or linear if every pair of elements is comparable.

Definition 15.3 (Well Order). A total order is a well order if every nonempty subset has a least element.

$(\mathbb{N}, \leq)$ is well ordered, but $(\mathbb{Z}, \leq)$ is not because $\mathbb{Z}$ has no least element.

15.3 Minimal, Maximal, Least and Greatest Elements

Let $S \subseteq P$.

- $m \in S$ is **minimal** if no distinct $x \in S$ satisfies $x \preceq m$.
- $m \in S$ is **maximal** if no distinct $x \in S$ satisfies $m \preceq x$.
- $l \in S$ is the **least** element if $l \preceq x$ for every $x \in S$.
- $g \in S$ is the **greatest** element if $x \preceq g$ for every $x \in S$.

A least or greatest element, if it exists, is unique. Minimal and maximal elements need not be unique.

15.4 Upper and Lower Bounds

For $S \subseteq P$:

- $u \in P$ is an upper bound if $x \preceq u$ for every $x \in S$.
- $l \in P$ is a lower bound if $l \preceq x$ for every $x \in S$.
- The least upper bound is denoted $\text{lub}(S)$ or $\sup S$.
- The greatest lower bound is denoted $\text{glb}(S)$ or $\inf S$.

Example 15.1. In $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$, for $A = \{1\}$ and $B = \{2\}$,

$$\text{lub}(A, B) = A \cup B = \{1, 2\},$$

$$\text{glb}(A, B) = A \cap B = \emptyset.$$

16 Hasse Diagrams

16.1 Purpose

A Hasse diagram is a simplified graphical representation of a finite poset. It omits reflexive loops, arrowheads and edges implied by transitivity.

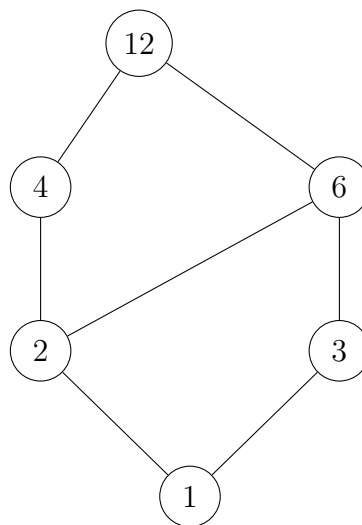
16.2 Construction Procedure

1. Verify that the relation is a partial order.
2. Draw one vertex for each element.
3. Place larger elements above smaller elements.
4. Connect x to y if y covers x .
5. Remove loops and all transitive edges.

Definition 16.1 (Cover). In a poset, y covers x if $x \prec y$ and there is no z such that $x \prec z \prec y$.

16.3 Example: Divisors of 12

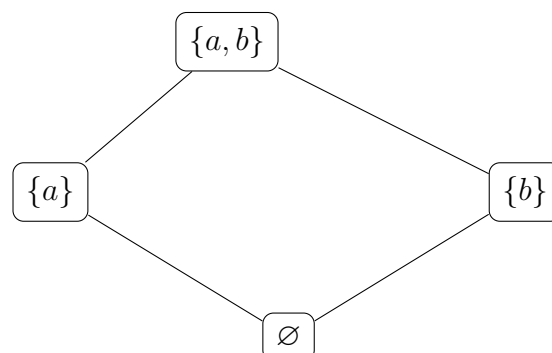
Let $P = \{1, 2, 3, 4, 6, 12\}$ ordered by divisibility.



Here 1 is the least element and 12 is the greatest element.

16.4 Example: Power Set

For $S = \{a, b\}$, the poset $(\mathcal{P}(S), \subseteq)$ has Hasse diagram



16.5 Reading Information from a Hasse Diagram

Moving upward means moving to a greater element. Immediate neighbours connected by an edge are cover-related. Common upper bounds lie above all selected elements; common lower bounds lie below them.

Common Mistake

Do not draw an edge for every comparable pair. A Hasse diagram contains only cover relations. Drawing transitive edges produces the relation digraph, not the Hasse diagram.

17 Lattices

Learning Outcomes

Students should be able to determine whether a poset is a lattice, calculate meet and join, and distinguish bounded, complemented, distributive, modular and complete lattices.

Definition 17.1 (Lattice). A poset $(L, \preceq)$ is a lattice if every pair $a, b \in L$ has a unique greatest lower bound and a unique least upper bound.

The meet and join are denoted

$$a \wedge b = \text{glb } \{a, b\}, \quad a \vee b = \text{lub } \{a, b\}.$$

17.1 Algebraic Laws of Lattices

For all $a, b, c \in L$:

$a \wedge a = a,$	$a \vee a = a$	(idempotent)
$a \wedge b = b \wedge a,$	$a \vee b = b \vee a$	(commutative)
$a \wedge (b \wedge c) = (a \wedge b) \wedge c,$	$a \vee (b \vee c) = (a \vee b) \vee c$	(associative)
$a \wedge (a \vee b) = a,$	$a \vee (a \wedge b) = a$	(absorption)

17.2 Examples of Lattices

1. $(\mathcal{P}(S), \subseteq)$ with meet $\cap$ and join $\cup$.
2. Positive divisors of n ordered by divisibility, with meet gcd and join lcm.
3. Any total order, where meet is minimum and join is maximum.

17.3 Bounded Lattice

Definition 17.2. A lattice is bounded if it has a least element 0 and greatest element 1 satisfying

$$0 \preceq a \preceq 1 \quad \forall a \in L.$$

17.4 Complemented Lattice

In a bounded lattice, b is a complement of a if

$$a \wedge b = 0, \quad a \vee b = 1.$$

A bounded lattice is complemented if every element has at least one complement.

In $\mathcal{P}(S)$, the complement of A is the set-theoretic complement $S - A$.

17.5 Distributive Lattice

A lattice is distributive if

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$$

and equivalently

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c).$$

Every Boolean lattice is distributive.

17.6 Modular Lattice

A lattice is modular if, whenever $a \preceq c$,

$$a \vee (b \wedge c) = (a \vee b) \wedge c.$$

Every distributive lattice is modular, but not every modular lattice is distributive.

17.7 Complete Lattice

Definition 17.3. A lattice is complete if every subset $S \subseteq L$, including the empty subset and L itself, has both a least upper bound and a greatest lower bound.

Every finite lattice is complete. The power set lattice $(\mathcal{P}(S), \subseteq)$ is complete because arbitrary unions and intersections exist.

17.8 Hierarchy of Important Classes

$$\text{Distributive} \Rightarrow \text{Modular} \Rightarrow \text{Lattice}.$$

The reverse implications are false in general.

17.9 Testing Whether a Poset Is a Lattice

For every unordered pair (a, b) :

1. identify all common lower bounds and verify a unique greatest one;
2. identify all common upper bounds and verify a unique least one.

If even one pair fails, the poset is not a lattice.

17.10 Solved Example: Divisors of 30

Let $L = \{1, 2, 3, 5, 6, 10, 15, 30\}$ ordered by divisibility. For any $a, b \in L$,

$$a \wedge b = \gcd(a, b), \quad a \vee b = \text{lcm}(a, b).$$

Since both values remain divisors of 30, L is a lattice. It is bounded by 1 and 30.

17.11 Counterexample

Consider a poset with two incomparable minimal common upper bounds of a and b . Since no unique least upper bound exists, the poset is not a lattice.

17.12 Applications

Lattices are used in Boolean algebra, digital circuit design, type systems, program analysis, access-control models, formal concept analysis, database dependency theory and fixed-point semantics.

18 Lexicographic Ordering

Learning Outcomes

Students should be able to compare tuples and strings using lexicographic ordering and explain its use in sorting, indexing and dictionary order.

Definition 18.1. Suppose $(A, \leq_A)$ and $(B, \leq_B)$ are ordered sets. The lexicographic order on $A \times B$ is defined by

$$(a_1, b_1) <_{\text{lex}} (a_2, b_2)$$

if either

$$a_1 <_A a_2,$$

or

$$a_1 = a_2 \text{ and } b_1 <_B b_2.$$

Thus the first coordinate has priority; later coordinates are compared only when earlier coordinates tie.

Example 18.1. Under the usual order on integers,

$$(2, 100) <_{\text{lex}} (3, 0)$$

because $2 < 3$, even though $100 > 0$.

18.1 Strings

Dictionary order is lexicographic order on strings. Compare characters from left to right. At the first mismatch, the string with the smaller character comes first. If one string is a prefix of another, the shorter string comes first.

Example 18.2.

$$\text{car} < \text{card} < \text{care} < \text{dog}.$$

18.2 Applications

Lexicographic ordering is used in database composite keys, sorting tuples, file-system ordering, dictionary implementation, combinatorial generation and version comparison.

19 Integrated Solved Problems

Example 19.1. Let $A = \{1, 2, 3, 4\}$ and

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1), (3, 4), (4, 3)\}.$$

Determine the type of R and its partition.

Solution. The relation is reflexive. Every non-diagonal pair occurs with its reverse, so it is symmetric. Its blocks are $\{1, 2\}$ and $\{3, 4\}$, and there are no cross-block pairs; therefore transitivity holds. Hence R is an equivalence relation, inducing partition

$$\{\{1, 2\}, \{3, 4\}\}.$$

Example 19.2. Let $P = \{1, 2, 3, 6, 12\}$ under divisibility. Find the minimal, maximal, least and greatest elements.

Solution. The least element is 1. The greatest element is 12. Therefore the only minimal element is 1 and the only maximal element is 12.

Example 19.3. In the divisor lattice of 60, compute $12 \wedge 20$ and $12 \vee 20$.

$$12 \wedge 20 = \gcd(12, 20) = 4,$$

$$12 \vee 20 = \text{lcm}(12, 20) = 60.$$

Example 19.4. Let $R = \{(1, 2), (2, 3), (3, 4)\}$. List R^2 and R^3 .

$$R^2 = \{(1, 3), (2, 4)\}, \quad R^3 = \{(1, 4)\}.$$

Thus

$$R^+ = \{(1, 2), (2, 3), (3, 4), (1, 3), (2, 4), (1, 4)\}.$$

20 Common Errors and Correction Strategy

Error	Why it is wrong	Correction
Confusing antisymmetric with asymmetric	Antisymmetric allows (a, a) and may allow one-way pairs	Apply the formal implication involving both directions
Treating membership as probability	Membership measures degree of belonging, not uncertainty of occurrence	Interpret the semantic meaning of the function
Including transitive edges in a Hasse diagram	Hasse diagrams display only cover relations	Remove every edge implied by a longer upward path
Assuming every poset is a lattice	A lattice requires unique meet and join for every pair	Test all pairs, not only obvious pairs
Writing $R \circ S$ in the wrong order	Composition order follows right-to-left application	Trace an element through the first relation, then the second
Calling maximal element greatest	Maximal need not dominate every element	Check whether it is above all elements
Skipping intermediate Warshall matrices	Final matrix alone may hide errors and lose marks	Show $W^{(0)}$ through $W^{(n)}$

21 Practice Questions

21.1 Multiple-Choice Questions

1. A fuzzy membership function maps a universe into:
 - (a) $\mathbb{N}$
 - (b) $\{0, 1\}$ only
 - (c) $[0, 1]$
 - (d) $\mathbb{Z}$
2. In a multiset, the union multiplicity is usually defined using:
 - (a) minimum
 - (b) maximum
 - (c) sum only
 - (d) subtraction
3. A relation that is reflexive, symmetric and transitive is:
 - (a) partial order
 - (b) equivalence relation
 - (c) asymmetric relation
 - (d) universal relation only
4. A partial order must be:
 - (a) reflexive, symmetric, transitive
 - (b) irreflexive, asymmetric, transitive
 - (c) reflexive, antisymmetric, transitive
 - (d) symmetric and antisymmetric only
5. Warshall's algorithm computes:
 - (a) shortest paths with weights
 - (b) minimum spanning tree
 - (c) transitive closure
 - (d) topological ordering
6. The time complexity of Warshall's algorithm is:
 - (a) $O(n)$
 - (b) $O(n^2)$
 - (c) $O(n^3)$
 - (d) $O(2^n)$
7. In a Hasse diagram, which edges are omitted?
 - (a) cover edges

- (b) transitive edges
 - (c) all upward edges
 - (d) all incomparable pairs
8. Every distributive lattice is:
- (a) modular
 - (b) complemented
 - (c) Boolean
 - (d) infinite
9. In a total order:
- (a) no elements are comparable
 - (b) every pair is comparable
 - (c) only adjacent elements are comparable
 - (d) every element is maximal
10. In the divisor lattice, meet and join are:
- (a) sum and product
 - (b) gcd and lcm
 - (c) min and max only
 - (d) difference and quotient

21.2 Short-Answer Questions

1. Define support and core of a fuzzy set.
2. Distinguish a multiset from a fuzzy set.
3. Give an example of a relation that is symmetric but not transitive.
4. Give an example of a relation that is antisymmetric but not asymmetric.
5. Define composition of two relations.
6. What is the quotient set of an equivalence relation?
7. Define reflexive closure and symmetric closure.
8. State the recurrence used in Warshall's algorithm.
9. Distinguish least element and minimal element.
10. Define a complete lattice.

21.3 Long-Answer and Numerical Questions

1. For given fuzzy sets A and B , compute union, intersection, complements and selected α -cuts.
2. Prove that congruence modulo n is an equivalence relation.
3. Construct the equivalence relation induced by a given partition.
4. Find reflexive, symmetric and transitive closures of a finite relation.
5. Apply Warshall's algorithm to a 4×4 matrix and show every stage.
6. Draw the Hasse diagram for the divisors of 24 under divisibility.
7. Draw the Hasse diagram of $\mathcal{P}(\{a, b, c\})$ under inclusion.
8. Identify all minimal, maximal, least and greatest elements in a given Hasse diagram.
9. Determine whether a given finite poset is a lattice.
10. Prove that every total order is a lattice.
11. Show that every distributive lattice is modular.
12. Compare modular and distributive lattices using examples.
13. Explain lexicographic order on ordered triples and sort a given list.

Answer Key for MCQs

1	2	3	4	5	6	7	8	9	10
(c)	(b)	(b)	(c)	(c)	(c)	(b)	(a)	(b)	(b)

22 Rapid Revision Sheet

- Fuzzy union: $\max$.
- Fuzzy intersection: $\min$.
- Fuzzy complement: $1 - \mu$.
- Multiset union: max multiplicity.
- Multiset sum: add multiplicities.
- Equivalence: reflexive + symmetric + transitive.
- Partial order: reflexive + antisymmetric + transitive.
- Reflexive closure: $R \cup I$.
- Symmetric closure: $R \cup R^{-1}$.
- Transitive closure: $R \cup R^2 \cup \dots$.
- Warshall complexity: $\Theta(n^3)$.
- Hasse diagram: remove loops, arrowheads and transitive edges.
- Lattice: every pair has unique meet and join.
- Bounded lattice: has 0 and 1.
- Complement: $a \wedge b = 0$, $a \vee b = 1$.
- Distributive $\Rightarrow$ modular.
- Complete lattice: every subset has sup and inf.
- Lexicographic order: compare first differing coordinate.